Mind Mirror: An AI-Powered Emotion-Aware Journaling Application

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***Abstract*— This paper presents Mind Mirror, an AI-powered journaling system designed to enhance emotional awareness through sentiment analysis, mood prediction, and visual trend tracking. Traditional journaling tools support self-expression but lack structured emotional analytics and integration with guided therapeutic workflows. Mind Mirror addresses this limitation by enabling users to record journal entries, receive automated sentiment classification, and visualize mood patterns over time through an interactive dashboard. The system further supports optional therapist collaboration by allowing selective sharing of entries. The platform is implemented using the FARM stack, consisting of a React frontend, a FastAPI-based Python backend, and MongoDB for data persistence, with a fine-tuned DistilBERT model used for mood prediction. System validation was conducted using synthetic and sample data, demonstrating successful functionality across all core features. The results demonstrate the feasibility of integrating AI-driven analysis with reflective journaling to support mental well-being.**

Keywords—artificial intelligence, sentiment analysis, journaling systems, digital mental health, natural language processing

I. Introduction

Emotional awareness is a central component of mental well-being, yet many individuals struggle to identify, interpret, and respond to their emotional states in a consistent manner. Traditional journaling provides a valuable outlet for self-expression, but it is typically unstructured and offers limited support for recognizing emotional patterns over time. As a result, many users record reflections without receiving meaningful analytical feedback or understanding how their emotional states evolve across days and weeks.

Recent advances in artificial intelligence, particularly in natural language processing (NLP), create opportunities to strengthen journaling tools through automated analysis of textual reflections. Sentiment analysis models can classify emotional tone, while dashboard-based visualizations can reveal longitudinal patterns that are difficult to notice in isolated entries. Prior work on expressive writing suggests that writing about personal experiences can contribute to improved emotional processing and health outcomes [3], [5]. In parallel, research on digital mental health applications suggests that accessible software tools can broaden support and increase user engagement when they are structured and easy to use [4].

This study introduces Mind Mirror, a web-based emotion-aware journaling application designed to connect reflective writing with AI-generated emotional insight. The platform enables users to submit journal entries, receive sentiment and mood-related feedback, and review their emotional history through visual dashboards. In its broader design, the platform also supports selective sharing of entries with therapists so that clinicians can review user-authorized content within a secure workflow.

The study addresses three research questions: How can sentiment analysis be integrated into a journaling system to enhance emotional awareness? How can trend visualization support pattern recognition and sustained reflection? How can a journaling platform support future therapist collaboration while preserving user control over shared content? Mind Mirror is intended to demonstrate that AI-assisted journaling can move beyond passive storage and toward structured, actionable self-reflection.

II. Literature Review

A. Expressive Writing and Emotional Health

Expressive writing has been widely studied as a means of improving emotional processing and psychological well-being. Pennebaker and Chung describe expressive writing as a structured activity through which individuals write about difficult experiences, often producing measurable benefits in health and well-being [3]. Smyth likewise reports that written emotional expression is associated with improved outcomes across multiple health-related dimensions, including psychological well-being and general functioning [5]. These findings establish a strong theoretical basis for systems that encourage reflective writing as part of mental wellness support.

B. Digital Mental Health Applications

Digital mental health platforms have expanded access to self-monitoring, reflection, and treatment-oriented support. A systematic review by Miralles et al. found growing interest in mobile applications for mental disorders, particularly for anxiety- and depression-related use cases, while also emphasizing the importance of sound technical and psychological design [4]. These findings support the usefulness of journaling systems that combine accessibility with guided structure instead of acting as plain text repositories.

C. Sentiment Analysis and Emotion Detection

Sentiment analysis methods provide a practical foundation for extracting emotional signals from user-written reflections. Hutto and Gilbert introduced VADER as an efficient, rule-based model for sentiment analysis of social media text and demonstrated that lightweight sentiment models can perform well on short, user-generated content [1]. Although journaling entries may be more nuanced than short social posts, the same principle applies: NLP can convert free-form text into structured emotional signals that support visualization and feedback.

D. Therapist-Supported Digital Systems

Digital self-monitoring systems can be strengthened when they support a path toward clinician involvement. de Zwaan et al. examined a web-based self-help and feedback intervention and found that automated monitoring systems can deliver meaningful support, while therapist involvement can improve user satisfaction even when the automated core remains central [2]. This supports Mind Mirror's long-term design goal of allowing users to selectively share entries with therapists for more contextual and personalized discussion.

III. METHODOLY

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A. Achitecture Diagram

Figure 1 illustrates the high-level architecture of Mind Mirror, which follows a client-server model implemented using the FARM stack. The frontend is developed using React and communicates with a FastAPI-based backend implemented in Python through RESTful APIs. The backend processes journal entries, interfaces with the sentiment analysis model, and manages data persistence using MongoDB. This architecture enables efficient handling of asynchronous requests and seamless integration with machine learning components. FastAPI provides high-performance API execution and native compatibility with Python-based NLP models.

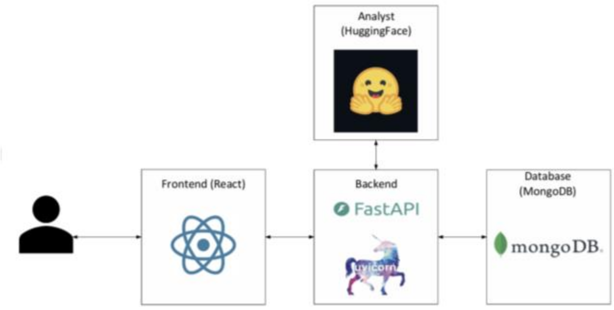

Fig. 1. Achitecture Diagram

B. ER Diagram

The data model shown in Figure 2 captures the primary entities used by the prototype. A single User can create many JournalEntry records, and each entry stores reflection text together with mood and sentiment fields. Each JournalEntry is associated with one Analysis record containing model metadata, sentiment labels, confidence information, and processed timestamps. A Suggestion record can optionally be linked to an entry to store advice text derived from analysis results. This schema supports separation between raw user reflections, model outputs, and optional feedback artifacts.

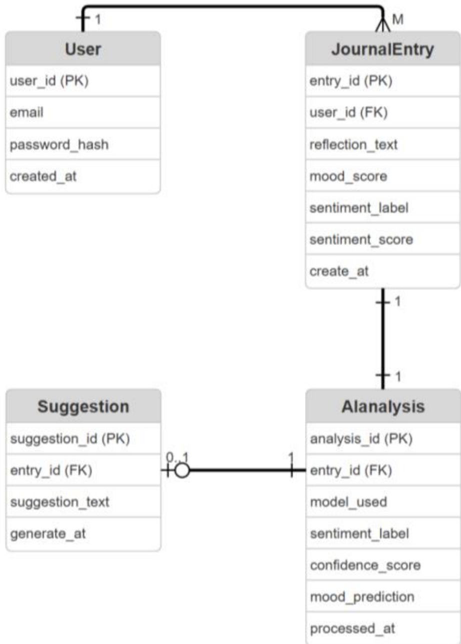

Fig. 2. ER Diagram

C. Context Diagram

Figure 3 presents the system context. The user writes a reflection through the journal input screen, submits the entry to Mind Mirror, and receives both a sentiment result and advice generated through AI-connected services. Mind Mirror stores and retrieves entries from MongoDB and sends prior records to the mood history and feedback view for longitudinal review. The context diagram makes clear that the system is not limited to one transaction; instead, it supports an ongoing cycle of entry, analysis, storage, and historical review.

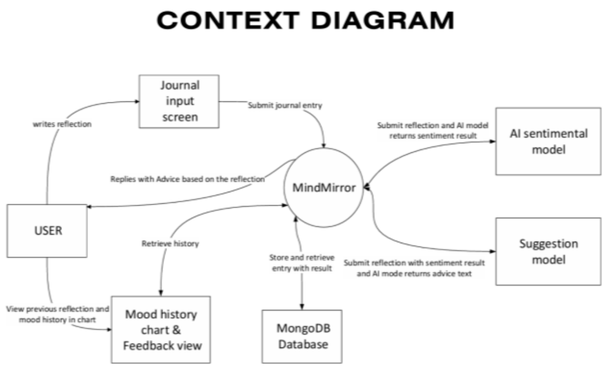

Fig. 3. Context Diagram

D. Sequence Diagram Journal Entry

The journal-entry workflow in Figure 4 begins when the user writes a reflection and submits it through the React frontend. The frontend issues a POST request to the FastAPI backend, which validates the payload and forwards the reflection text to the sentiment and emotion model. The model returns structured outputs such as sentiment label, confidence score, emotion label, and mood score. The FastAPI backend then stores both the original entry and the analysis result in MongoDB before returning the processed data to the frontend for display. This sequence supports immediate feedback while preserving a durable analysis history.

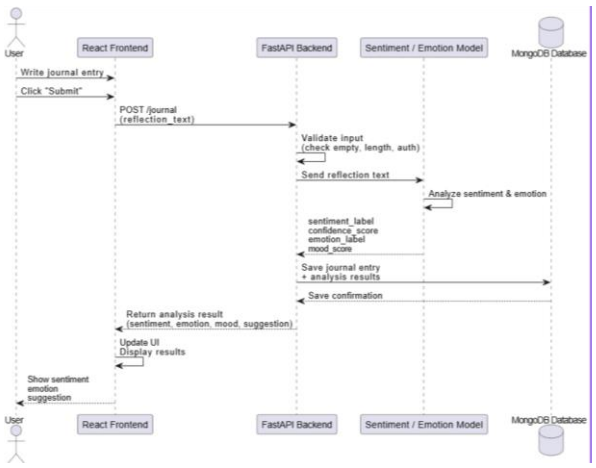

Fig. 4. Sequence Diagram – Journal Entry

E. Sequence Diagram Dashboard

Figure 5 shows the dashboard workflow. When the user opens the dashboard, the React frontend requests journal history from the FastAPI backend. The FastAPI backend queries MongoDB for entries associated with the current user, formats the returned records into a JSON payload, and sends them back to the frontend. The frontend then generates a mood chart and entry list so that the user can examine mood trends and prior reflections. This workflow is essential to the system's value because emotional pattern recognition depends on reviewing multiple entries over time rather than analyzing one entry in isolation.

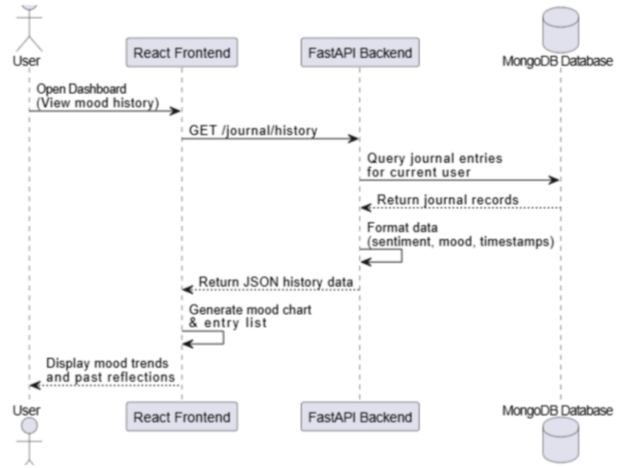

Fig. 5. Sequence Diagram – Dashboard

F. Sequence Diagram Dashboard

Mind Mirror incorporates a fine-tuned natural language processing model to generate sentiment classifications and continuous mood scores from journal entries. The model is based on the DistilBERT architecture and is implemented using the HuggingFace Transformers framework with a PyTorch backend.

The model utilizes the DistilBERT-base-uncased encoder, which consists of six transformer layers, a hidden size of 768, and 12 attention heads, totaling approximately 66 million parameters. A linear regression head is applied to the encoded representation to predict a numerical mood score on a scale from 1 (very negative) to 5 (very positive).

The model is trained using the GoEmotions dataset, which contains approximately 58,000 labeled text samples. The dataset is divided into 43,410 training samples, 5,426 validation samples, and 5,427 test samples. Each sample includes one or more emotion labels from a set of 27 categories.

To convert multi-label emotion annotations into a single numerical value, a mapping scheme is applied. Positive emotions such as joy, love, gratitude, and optimism are mapped to higher scores (5), while negative emotions such as anger, fear, and sadness are mapped to lower scores (1). Neutral or ambiguous emotions are assigned mid-range values. The final mood score is computed as the average of all mapped values for each sample, allowing transformation from categorical labels to a continuous regression target.

Text inputs are tokenized using the DistilBERT tokenizer with truncation and padding applied to match model input requirements. Non-essential fields are removed, and the processed dataset is used to train the regression model.

Training is conducted using the HuggingFace Trainer API with the following configuration: two training epochs, a batch size of 16, and a learning rate of 2×10^{-5}. The AdamW optimizer is used with weight decay. Model performance is evaluated using Mean Squared Error (MSE) and Root Mean Squared Error (RMSE), which are standard metrics for regression tasks.

During runtime, journal entries are transmitted from the React frontend to the FastAPI backend, which processes inference requests using the trained model. The backend returns both sentiment classifications and mood scores, which are stored in MongoDB and used for visualization within the dashboard.

Although effective, the model has several limitations. The mapping from categorical emotions to numerical mood values may oversimplify complex emotional states. Additionally, the model is trained on Reddit-based data, which may not fully generalize to personal journaling contexts. The regression output is also not strictly bounded, and extreme inputs may produce values outside the intended range.

G. Authentication and Role-Based Access Control

The system implements a stateless authentication mechanism using JSON Web Tokens (JWT). Upon successful login, the backend generates a signed token containing the user ID and role, which is included in subsequent API requests via the Authorization header. Token expiration is configurable and ensures session security.

Two roles are defined: patient and therapist. Role-based access control is enforced through backend validation layers, restricting access to endpoints based on user permissions. Patients can create and manage journal entries, while therapists are limited to read-only access for authorized patient data. Additional middleware ensures that therapist accounts cannot perform data mutation operations.

H. Therapist Collaboration Model

The implemented system introduces a three-layer access control model to preserve user privacy:

1. Link Layer: Patients must explicitly link to a therapist via a unique therapist ID.

2. Global Sharing Toggle: A system-wide setting determines whether any data can be shared.

3. Per-Entry Sharing: Each journal entry includes a boolean flag controlling therapist visibility.

Only when all three conditions are satisfied can a therapist access a specific journal entry. This ensures granular control and aligns with ethical considerations in digital mental health systems.

I. Extended Data Model

The system extends the original data model by adding new fields directly to the JournalEntry structure, including:

- emotion_label
- emotion_confidence
- predicted_mood
- sentiment_confidence
- shared_with_therapist
- suggestion

Additionally, two new collections are introduced:

- therapist_patient_links (relationship mapping)
- patient_profiles (sharing configuration)

J. Pipeline Enhancements

Beyond sentiment analysis and mood prediction, the system integrates:

- Emotion classification using a HuggingFace model
- Suggestion generation, implemented via:
    - o Rule-based logic (default)
    - o Optional LLM-based generation using Groq API

Suggestions are dynamically generated based on emotional context and displayed with color-coded visual feedback.

IV. RESULTS

Mind Mirror was successfully implemented as a functional prototype. In a local development environment, the system supported journal submission, automated sentiment analysis, dashboard-based history retrieval, and data persistence. The frontend remained responsive during common interactions, and the API workflow consistently returned structured results for display. These outcomes indicate that the proposed architecture is feasible for a real-time journaling application in which text analysis and user feedback occur within the same interaction cycle.

The prototype also demonstrated utility at the interface level. Users can write reflections in free-form language, receive immediate sentiment-related feedback, and inspect prior entries through a chart-oriented dashboard. This combination of immediate analysis and longitudinal review helps transform journaling from a passive activity into a guided reflective process. Literature on expressive writing and digital mental health suggests that this structure can support emotional awareness, stress reduction, and sustained engagement when fully evaluated with users [3]-[5].

An important design observation is that sentiment analysis alone is not sufficient unless results are retained and visualized over time. The dashboard workflow therefore serves as a critical complement to the journal-entry pipeline. In the same way, optional suggestion generation adds value by translating analytical output into user-facing guidance. Together, these functions demonstrate how a journaling tool can offer structured support without replacing human judgment or professional care.

The extended system was evaluated through end-to-end workflow testing. Therapist collaboration features were successfully validated, including account provisioning, patient linking, controlled data sharing, and dashboard visualization.

A. PER ENTRY SHARING FUNCTIONALITY

Per-entry sharing functionality was validated through API-level and interface-level testing. Toggle operations using the PATCH /api/checkins/{id}/sharing endpoint successfully updated the shared_with_therapist field in MongoDB. The backend correctly resolved both ObjectId and string-based identifiers, ensuring compatibility with entries created across different development stages.

The frontend implemented optimistic UI updates, immediately reflecting toggle changes while maintaining consistency through rollback mechanisms in case of API failure. This behavior ensured both responsiveness and data integrity.

B. Therapist workflow End-to-End Validation

The complete therapist collaboration workflow was validated using controlled test scenarios. An administrator successfully provisioned a therapist account through the protected endpoint, with invalid authentication attempts correctly rejected.

A patient then linked the therapist account, enabled global sharing, and selectively shared a subset of journal entries. Upon login, the therapist dashboard correctly displayed the linked patient. Only entries explicitly marked for sharing were visible, while private entries remained inaccessible.

Mood trend visualization rendered correctly based on shared entries. When global sharing was disabled by the patient, subsequent therapist access was restricted, confirming enforcement of the multi-layer access control model. These results validate the correctness of the end-to-end workflow and integration between frontend, backend, and database components.

C. Data integrity and Orphan Handling

System robustness was evaluated through orphaned data scenarios. After manually removing a user record from the database, associated references in linking collections were tested. The system correctly excluded invalid references from therapist views, preventing display of non-existent users.

Additionally, cleanup scripts successfully removed orphaned records, ensuring long-term data consistency. This demonstrates that the system can maintain integrity even under abnormal data conditions.

D. Suggestion System and visual Feedback

The suggestion generation system was evaluated through both rule-based and LLM-assisted modes. Suggestions were successfully generated based on sentiment and emotion outputs and displayed to users in real time.

A color-coded visualization scheme was consistently applied, with positive, neutral, and negative sentiments mapped to distinct visual styles. This enhanced interpretability and user engagement without affecting system performance. When LLM integration was enabled, generated suggestions replaced rule-based outputs while preserving visual consistency.

V. THERAPIST COLLABORATION WORKFLOW

The system includes a fully implemented therapist collaboration feature. Therapists are provisioned through an admin-controlled endpoint and can access patient data only when explicitly authorized.

Patients can link therapists, enable sharing, and selectively expose journal entries. Therapists access this data through a dedicated dashboard that provides:

- Patient summaries
- Journal entries (read-only)
- Mood trend visualizations

Access is strictly enforced through backend validation to ensure privacy and compliance with ethical design principles.

VI. CONCLUSION

Mind Mirror demonstrates how AI-assisted sentiment analysis can be integrated into a journaling application to promote emotional awareness and reflective insight. By combining journal entry analysis with historical visualization, the system provides a more structured alternative to traditional digital diaries. The architecture, data model, and workflow diagrams show that the design can support real-time analysis, durable storage, and trend-based review within a cohesive web application.

The present study has several limitations. Formal user testing was not conducted during this phase, and evaluation was limited to prototype functionality and literature-supported expectations. The system has been validated in a development environment, but performance benchmarking in production remains incomplete. Sentiment and emotion models may oversimplify complex psychological states, and mapping categorical emotions to continuous mood scores may reduce nuance. Additionally, while the therapist-sharing workflow is fully implemented, it has not yet undergone clinical validation or formal security auditing required for healthcare deployment.

The therapist-sharing workflow described in the initial design has now been fully implemented, including granular per-entry sharing, a therapist dashboard, and secure access controls. Future work will focus on empirical user studies, mobile optimization, enhanced therapist interaction features, compliance with healthcare data standards, and domain-specific model fine-tuning to improve emotional accuracy.

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APPENDIX A – API ENDPOINTS

The table below documents all endpoints implemented in the system, grouped by domain.

A. Authentication Endpoints

Method	Path	Description
POST	/api/register	Patient self-registration. Role hardcoded to patient.
POST	/api/login	Authenticate any user. Returns JWT + id + role.
GET	/api/me	Return current user's profile, id, and role.
POST	/api/admin/therapists	Admin-only. Create a therapist account. Requires X-Admin-Key header.

B. Check-in (Journal Entry) Endpoints

Method	Path	Description
POST	/api/checkin	Submit a journal entry. Triggers NLP analysis synchronously. Accepts shared_with_therapist flag.
GET	/api/checkins	Retrieve all check-ins for the authenticated patient.
PATCH	/api/checkins/{id}/sharing	Toggle shared_with_therapist for a specific entry. Patient-only.
GET	/api/insights	Retrieve insights summary, 7-day averages, and chart data for the patient.
GET	/api/past-entries	Retrieve paginated past entries with full metadata.
GET	/api/mood-history	Retrieve sentiment + confidence time series for the patient.

C. Sharing Settings Endpoints

Method	Path	Description
GET	/api/sharing/status	Return all therapist links and global sharing flag for the patient.
PATCH	/api/sharing/enabled	Toggle the patient's global sharing_enabled flag.
POST	/api/sharing/link	Link a therapist to the patient by therapist ID.
DELETE	/api/sharing/link/{therapist_id}	Soft-delete (deactivate) a therapist link.

D. Therapist Data-Access Endpoints

Method	Path	Description
GET	/api/therapist/patients	List all linked, active patients for the therapist.
GET	/api/therapist/patients/{id}/profile	Get name, email, sharing status of one patient.
GET	/api/therapist/patients/{id}/journal	Get shared journal entries for one patient. Enforces three-layer access check.
GET	/api/therapist/patients/{id}/mood	Get sentiment trend data for one patient. Enforces three-layer access check.

All endpoints are protected using JWT-based authentication and enforce role-based access control to ensure secure data handling.